

# Assignment-1

## **1. What is Full Stack Development? Explain the term Full Stack Development and the role of a Full Stack Developer.**

- Full Stack Development means building both the frontend and backend of a website. A Full Stack Developer works on the design, logic, and database together.

## **2. What is Frontend? Explain Frontend and list some technologies used to build it.**

- Frontend is the part of the website that users see and interact with. Examples: HTML, CSS, JavaScript.

## **3. What is a Database? Why do websites need databases? Give examples of data stored in a database.**

- Database stores information for the website. Websites need it to save user details, products, or records. Example: student names, product prices.

## **4. What is DNS? Explain how DNS helps users access websites.**

- DNS changes website names into IP addresses. It helps users open websites easily without remembering numbers.

## **5. What is an IP Address? Why are IP addresses important in networking?**

- IP Address is a unique number for every device on the internet. It is important for communication between computers.

## **6. What is Client-Server Architecture? Explain how a client and server communicate with each other.**

- Client (browser) sends a request, and Server replies with data. This is how they talk to each other

## **7. What is an API? Explain the role of APIs in web applications.**

- API connects two applications. Example: Payment API connects a shopping site with a bank.

## **8. What are HTTP Methods? Explain the purpose of: GET • POST • PUT • DELETE**

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## GET

Used to retrieve data from a server.

Example: Viewing products on Amazon.

## POST

Used to send new data to a server.

Example: Registering a new account.

## PUT

Used to update existing data.

Example: Updating profile information.

## DELETE

Used to remove data.

Example: Deleting a user account.

## 9. What are Developer Tools? Name any three tabs available in Developer Tools and explain their purpose.

- Developer Tools are browser tools that help developers inspect, test, and fix websites. They can be opened by pressing F12.
  - Three Tabs in Developer Tools
- 1. Elements Tab
  - Shows the HTML and CSS of a webpage.
  - Used to inspect page structure and design.
- 2. Network Tab
  - Shows all files loaded by the website.
  - Helps check website loading speed.
- 3. Console Tab
  - Shows errors and messages.
  - Used for testing and debugging JavaScript code.

## Part B: Diagram Activities

### 10. Draw the Full Stack Architecture Diagram. User ↓ Frontend ↓ Backend ↓ Database Write a short description of each layer.

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- **User:** Person who uses the website.
- **Frontend:** User interface visible on screen.
- **Backend:** Processes requests and business logic.
- **Database:** Stores application data permanently.

## 11. Draw the Complete Website Working Flow. User ↓ Browser

↓ DNS ↓ Server ↓ Database ↓ Server ↓ Browser ↓ Website  
Displayed Explain each step in one sentence.

- User enters a website URL.
- Browser receives the request.
- DNS converts the domain name into an IP address.
- Browser connects to the server.
- Server requests data from the database.
- Database sends data to the server.
- Server sends the response to the browser.
- Browser displays the webpage to the user.

## • Part C: Real-World Application Questions.

## 12.. Student Registration System A student fills a registration form on a website. Explain: • What happens in the Frontend? • What happens in the Backend? • What happens in the Database?

### ➤ Frontend

- The student fills in the registration form and clicks the submit button.

### ➤ Backend

- The backend receives the submitted information, checks whether it is valid, and processes it.

### ➤ Database

- The student's details are stored permanently in the database.

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**13. Amazon Search Scenario** A user searches for "Laptop" on Amazon. Explain the complete flow from entering the search term to displaying the results. Use the following terms in your answer: • **Browser** • **Server** • **Database** • **Response**

- The user enters "Laptop" in the Amazon search bar using a **Browser**.
- The browser sends the search request to the **Server**.
- The server searches for matching products in the **Database**.
- The database returns the product information to the server.
- The server prepares a **Response** containing the search results.
- The response is sent back to the browser.
- The browser displays the list of laptops on the screen.

**14. Website Observation Activity** Open any website and inspect it using Developer Tools (F12). Write: • **Website Name** • **One observation from the Elements Tab** • **One observation from the Network Tab** • **One observation from the Console Tab**

- **Website Name:**
- Google
- **Elements Tab Observation:**
- The page structure is created using HTML elements like div, input, and button tags.
- **Network Tab Observation:**
- Many files such as HTML, CSS, JavaScript, and images are loaded when the page opens.
- **Console Tab Observation:**
- The console shows messages and errors generated by JavaScript.

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## Reflection Questions

**15.. In your own words, explain how a website works from the moment a user enters a URL until the webpage appears on the screen.**

➤ When a user enters a URL in the browser, the browser asks DNS for the website's IP address. After getting the IP address, the browser connects to the server. The server processes the request and fetches required data from the database. The server then sends a response back to the browser, which displays the webpage on the user's screen.

**16.Explain the difference between Frontend, Backend, and Database using a real-world example of your choice (Amazon, Instagram, Zomato, Netflix, etc.).**

- Frontend
- The frontend is what users see, such as the Amazon homepage, product images, search bar, and buttons.
- Backend
- The backend processes user actions like searching for products, logging in, and placing orders.
- Database
- The database stores product details, customer information, order history, and payment records.
- Simple Example:
  - When you search for a laptop on Amazon:
- Frontend shows the search bar and results.
- Backend processes your search request.
- Database stores all product information and sends it to the backend.
- This way, all three components work together to make the website function properly.